

ROUTE–RETRIEVE–VERIFY: Domain-Specialized Evidence Routing for Bengali Hallucination Detection

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অলীকবচন | Bengali LLM Hallucination Detection Challenge

Abstract

Bengali hallucination detection combines numerical verification, lexical and idiomatic interpretation, open-domain fact checking, and contextual entailment under one binary label. We present ROUTE–RETRIEVE–VERIFY, an offline pipeline that routes each row to one of four deterministic domain experts before selecting evidence. Arithmetic is recomputed, direct lexical questions use a typed archive, and only evidence-seeking rows invoke dense–sparse retrieval. FAISS candidates receive a sparse–dense preliminary rank, are cross-encoder reranked, and pass through a confidence filter before a shared Qwen3.5-35B-A3B GPTQ-Int4 verifier is called. Selective knowledge retrieval reduces distractors and token cost, while specialized relation queries recover synonym and antonym evidence that a single semantic query misses. The complete pipeline runs offline with vLLM on two Tesla T4 GPUs: 3 hours 6 minutes for 2,516 rows and 4 hours 59 minutes for the 5,000-row held-out evaluation.

1 Introduction

Bengali response verification is heterogeneous: a fluent answer can still contain a wrong calculation, a literal reading of an idiom, an entity confusion, or a contradiction of the supplied passage. These failures demand different evidence and decision rules. Bengali hallucination benchmarks also show substantial task-wise variation (Adib et al., 2026).

We therefore route before we retrieve or verify. A deterministic *domain expert router* assigns every row to numerical, Bengali linguistic, no-context factual, or context-grounded verification. These are prompt–evidence experts, not four models; one shared verifier retains semantic capacity while routing remains inexpensive, auditable, and complete. Figure 1 summarizes the deployed topology.

2 Method

2.1 Domain Expert Routing

An instance is $x = (c, q, r)$: optional context, Bengali prompt, and candidate response. The label $y \in \{0, 1\}$ denotes hallucinated or faithful. After shared Unicode,

Expert	Primary signal	Verification policy
Numerical	operation, quantity, pattern	derive final value
Linguistic	grammar or lexical intent	archive/relation evidence
No-context factual	neither above; $c = \emptyset$	retrieved fact or knowledge
Context-grounded	neither above; $c \neq \emptyset$	direct context, then RAG

Table 1: The four verification experts activated by the domain expert router.

quote, whitespace, and Bengali-digit normalization, the router applies a high-precision math check g_m , then a linguistic-intent check g_l , and finally a context-null split:

$$E(x) = \begin{cases} E_{\text{math}}, & g_m(q, r) = 1, \\ E_{\text{ling}}, & g_m = 0 \wedge g_l(q) = 1, \\ E_{\text{ctx}}, & g_m = g_l = 0 \wedge c \neq \emptyset, \\ E_{\text{fact}}, & \text{otherwise.} \end{cases} \quad (1)$$

The order is consequential: superficially numerical GK is not sent to a calculator, while quoted entities are not automatically treated as words whose meanings are requested.

The four outputs are asserted to be a one-to-one partition of official IDs; missing, duplicate, or extra assignments stop inference, and unmatched rows enter a general fallback. Specialization removes instruction interference and limits each row to evidence that can establish its answer. Unlike a learned router, the domain expert router has no training or inference cost and every decision is traceable to a rule.

2.2 Selective Knowledge Retrieval

Here, *selective* means that the system searches the knowledge base only when the assigned expert needs external evidence. Math bypasses RAG because retrieved prose cannot verify a fresh calculation. Direct word, literal, figurative, and idiom requests first use a typed Bengali archive indexed by normalized primary and alternative forms. Literal and figurative fields remain separate. General factual rows and context rows whose passage does not answer the question use ordinary RAG; synonym and

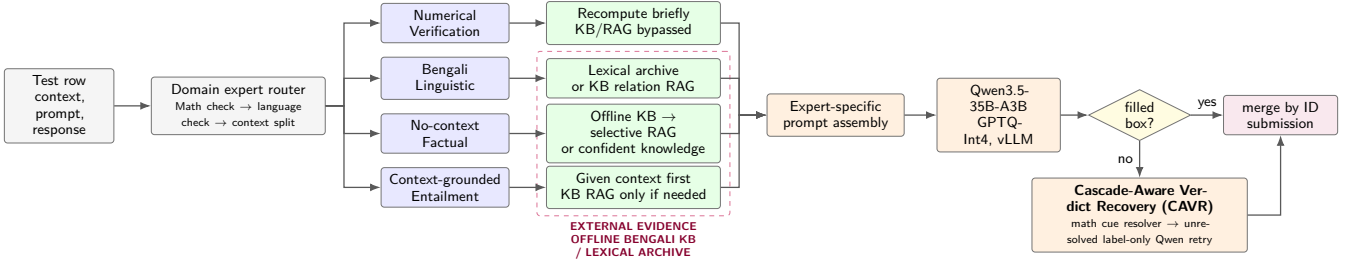


Figure 1: End-to-end inference architecture. The domain expert router selects one of four paths. Math bypasses retrieval; the other experts use the supplied context, lexical archive, or selective RAG over the offline Bengali KB as appropriate. The shared verifier emits a label; CAVR repairs missing filled-box verdicts.

antonym rows use a relation-specific path. Thus, retrieval is an expert tool, not mandatory decoration on every prompt.

Offline knowledge construction. Documents are normalized with NFKC, zero-width character removal, unified quotes and digits, and collapsed whitespace at both indexing and query time. Markup/navigation noise is removed, text is split at meaning boundaries, and near-duplicates are filtered before encoding. Meaning-aware chunks avoid cutting definitions from their headwords. Deduplication also protects the three-slot prompt budget: near-identical pages cannot occupy every evidence position and suppress source diversity. Clean text, dense vectors, sparse weights, and source metadata share a document ID; any alignment mismatch aborts retrieval.

Hybrid candidate scoring. BGE-M3 produces dense and sparse query representations (Chen et al., 2024). FAISS first obtains a dense candidate set $D_{30}(q)$ with $n_{\text{probe}} = 4096$. Sparse scores are then evaluated on the same candidates. Because the score scales differ, each is normalized within the query pool and fused for a preliminary ordering:

$$s_h(q, d) = \alpha \hat{s}_d(q, d) + (1 - \alpha) \hat{s}_s(q, d), \quad \alpha = 0.7, \quad (2)$$

where $\hat{s} = (s - \min_{D_{30}} s) / (\max_{D_{30}} s - \min_{D_{30}} s)$ (and is zero for a constant pool). The cross-encoder subsequently scores all 30 candidates, so final evidence order and acceptance are determined by s_{ce} ; we do not attribute an isolated gain to this preliminary fusion.

Relation-aware retrieval and abstention. A query asking whether candidate r is an antonym or synonym of target t often retrieves a definition of t , not evidence about the pair. We build $V(q, t, r)$ from the original prompt, a direct relation request, a canonical target–candidate statement, and a paired verification query. The candidate set is

$$C(q, t, r) = \bigcup_{v \in V(q, t, r)} \text{Top}_{10}(D_{30}(v), s_{\text{ce}}(v, d)), \quad (3)$$

deduplicated by document and reranked against the canonical relation query. Direct relation assertions outrank mere co-occurrence. For both ordinary and relation retrieval, the evidence shown to the verifier is

$$\mathcal{R}(q) = \text{Top}_3\{d \in C(q) : s_{\text{ce}}(q, d) > \tau\}, \quad \tau = 0.75. \quad (4)$$

The strict inequality matches the implementation. Importantly, $\mathcal{R}(q)$ may be empty: abstaining is safer than injecting a topical but wrong-entity passage.

2.3 Evidence Arbitration and Expert Prompts

Supplied context, RAG, and lexical evidence remain separate prompt blocks. The ordinary precedence is: direct answer in supplied context, relevant exact-entity RAG, then confident parametric knowledge. This preserves authoritative context without forcing an irrelevant passage to support an unrelated question. The typed lexical archive is consulted only for a matching term and requested relation or meaning type.

The numerical expert requests a brief derivation and final-value comparison; the linguistic expert checks the requested meaning type while accepting semantic equivalents. Factual prompts reject wrong entities but tolerate harmless abbreviations, date granularity, and rounding. Every prompt ends with `FINAL_LABEL=\boxed{0|1}`, followed by a deterministic parser. If the filled box is absent, our *Cascade-Aware Verdict Recovery* (CAVR) first scans mathematical outputs for unambiguous correctness or mismatch cues. A resolved cue produces the label directly; all remaining no-box cases receive a constrained, label-only Qwen retry. CAVR therefore recovers formatting failures without repeating retrieval or inviting another long reasoning chain.

The shared inference model is Qwen3.5-35B-A3B GPTQ-Int4 (Qwen Team, 2025; Frantar et al., 2023), served by vLLM (Kwon et al., 2023) with tensor parallelism on two T4 GPUs. Retrieval runs in a child process and exits before vLLM starts, fully releasing the embedding and reranker CUDA contexts. The offline run uses FP16 activations, eager execution, chunked prefill, prefix

Knowledge family	Representative sources	Raw format / unit	Raw size	Indexed units
Grammar, orthography, and lexicon	Samas, sandhi, prefixes, spelling, dictionaries, synonyms/antonyms, idioms	CSV/JSON/text; QA or typed entry	~1.5 GB	1,885,740 chunks
Law, administration, and civic services	bdlaws.gov.bd, Constitution, civil-service rules, Citizen Charter, district entities	JSON/JSONL/CSV; article, rule, or service	~127 MB	190,558 chunks + 45,321 rows
Curriculum, literature, and commerce	NCTB classes 6–12, SSC science/ICT, writer biographies, <i>Lalsalu</i> , accounting BCS/admission banks, current affairs, Bangla MMLU, cultural and diverse QA	PDF/CSV/Markdown; section or QA	≥9 MB	31,762 chunks + 5,317 rows
GK, MCQ, and culture	BanglaMMLU, cultural and diverse QA	CSV/JSON; QA or MCQ	~37 MB	7,385 chunks + 56,903 rows
Medical and bilingual QA	BanglaMMedBench, BanglaMedQA, English–Bangla medical QA and facts	CSV; QA or fact	~6 MB	9,413 rows
News and transcribed material	Prothom Alo article pipeline and 13 Saon transcript documents	JSON/Markdown; article or transcript	not consolidated prepared; not retained in final v4	

Table 2: Knowledge-source inventory compiled from the construction manifest, covering all 22 listed source types. Sizes can overlap across processing stages; “rows” denotes UNIFIED-pipeline inputs without a consolidated post-dedup count, and prepared but unindexed assets are marked explicitly.

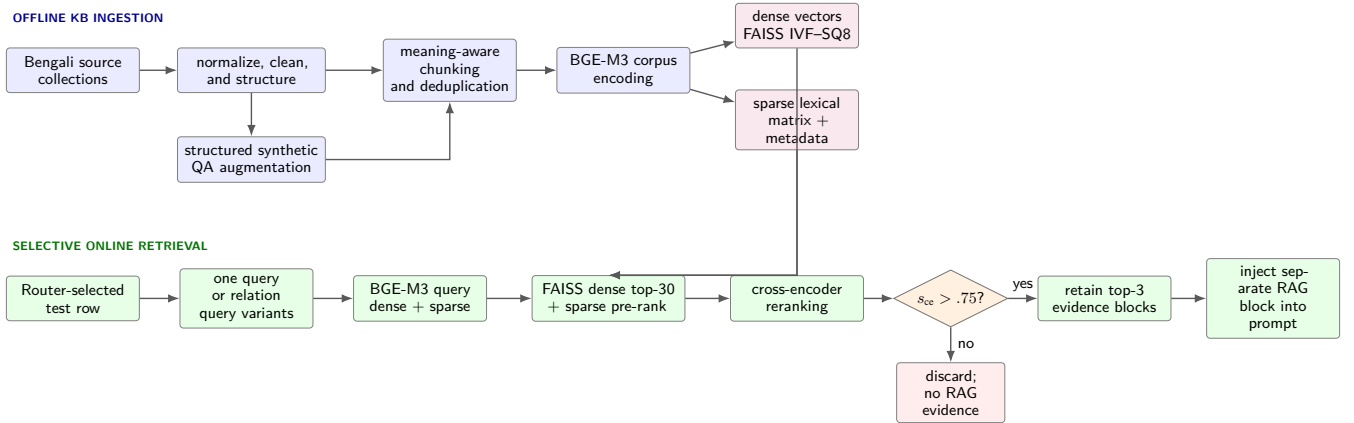


Figure 2: Knowledge ingestion and selective retrieval. Clean source text and structured QA augmentation are chunked before dual BGE-M3 indexing. At run time, ordinary factual rows use one query, while synonym/antonym rows use the multi-query relation path in Equation 3; only reranker scores above 0.75 enter the prompt.

caching, a 6,000-token context limit, 780 generation tokens, and repetition penalty 1.2. Careful vLLM scheduling, prompt-length control, tensor parallelism, and KV-cache allocation reduced the selected model’s inference pass from approximately 5 hours to 2 hours 18 minutes, a $2.17\times$ speedup, without changing the model or reducing its generation allowance.

3 Experiments and Results

We evaluate with macro-F1. Table 3 reports *raw model-pass results only*: direct inference before domain expert routing, archive lookup, RAG, and other pipeline components. The listed time is therefore model inference latency on the evaluation set, not end-to-end pipeline runtime.

Base model	Form	Raw F1	Inference time
Qwen3-4B-Instruct	standard	.6900	1 h 52 m
Qwen3-14B-Instruct	standard	.7110	2 h 48 m
Qwen3-30B-A3B	GPTQ-Int4	.7900	4 h 00 m
Qwen3.5-35B-A3B	GPTQ-Int4	.8224	2 h 18 m
Qwen3.6-35B-A3B	GPTQ-Int4	.8110	2 h 10 m

Table 3: Raw direct-inference comparison. F1 and time exclude retrieval, archive lookup, routing, and all other end-to-end pipeline stages.

Configuration	Evaluation	Macro-F1
Single general prompt	public LB	.8224
+ curated lexical archive	public LB	.8440
+ selective RAG	public LB	.8700
+ cross-encoder reranker	public LB	.8800
Final, updated KB without news	private LB	.9060
	public LB	.9140
	held-out	.92111
News-extended KB (separate run)	private LB	.9340
	public LB	.9410

Table 4: Incremental public-leaderboard results, followed by the final private-leaderboard, public-leaderboard, and held-out evaluations.

Protocol. The submitted merged notebook used the updated knowledge base without newspaper content and ran end to end in 3 h 06 m for 2,516 rows on two Tesla T4 GPUs. On the approximately 5,000-row held-out set, the same offline pipeline completed in 4 h 59 m, remaining within the nine-hour execution budget. Routing rules, archive matching, α , and τ were frozen before leaderboard evaluation. A fixed manually audited subset was retained across prompt revisions so retrieval and prompt effects could be separated from label noise. We also evaluated a newspaper-enriched index that was approximately twice the size of the submitted index. It improved both

leaderboard splits, but was not incorporated into the final merged notebook because its full offline integration and reproducibility could not be revalidated within the submission window. We therefore present those scores as a separate extension rather than as results of the officially submitted pipeline.

4 Error Analysis

Development errors mapped directly to design changes. Fluent but incorrect math motivated short recomputation; literal/figurative confusion motivated typed archive fields; irrelevant non-null passages motivated “context answers” rather than “context exists” priority; and wrong-entity retrieval motivated cross-encoder relevance filtering. Relation query expansion addressed a separate failure: high topical overlap with no explicit synonym/antonym assertion.

Observed risk	Naive behavior	Final safeguard
Wrong entity	trusts topical RAG	entity check + reranker filter
Literal/figurative	accepts plausible meaning	typed archive fields
Partial name/date	over-rejects answer	semantic equivalence rule
Fluent wrong math	trusts explanation	recompute final value

Table 5: Qualitative error-to-design traceability.

5 Reproducibility and Safeguards

Context, lexical archive, RAG, and model knowledge remain distinct sources. The routed policy uses direct context first, then an applicable typed lexical record or rele-

vant RAG, and finally confident model knowledge. The notebook also validates index–metadata alignment, one-to-one ID coverage, prompt budgets, binary parser output, and final submission order.

Generation is deterministic (temperature zero). Models, indexes, archives, router code, and Python wheels are mounted locally, so the submitted notebook contains no network dependency. The retrieval child process exits before the verifier loads, making GPU ownership sequential rather than concurrent.

6 Limitations

The deterministic domain expert router trades adaptability for transparency: unseen phrasing may reach the fallback expert. The global reranker threshold is not domain-calibrated, the archive is coverage-limited, and relation-query rules do not recognize every paraphrase. Quantization and dual-T4 serving constraints were not isolated from model quality.

7 Conclusion

ROUTE–RETRIEVE–VERIFY separates domain recognition, evidence acquisition, and final judgment. Its deterministic domain expert router activates four specialized verification policies while one quantized multilingual model supplies the shared semantic capacity. Selective knowledge retrieval and a curated lexical archive improve grounding without burdening every row with noisy evidence. The resulting design is reproducible offline, auditable at each stage, and compatible with the compute constraints of dual-T4 Kaggle inference.

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